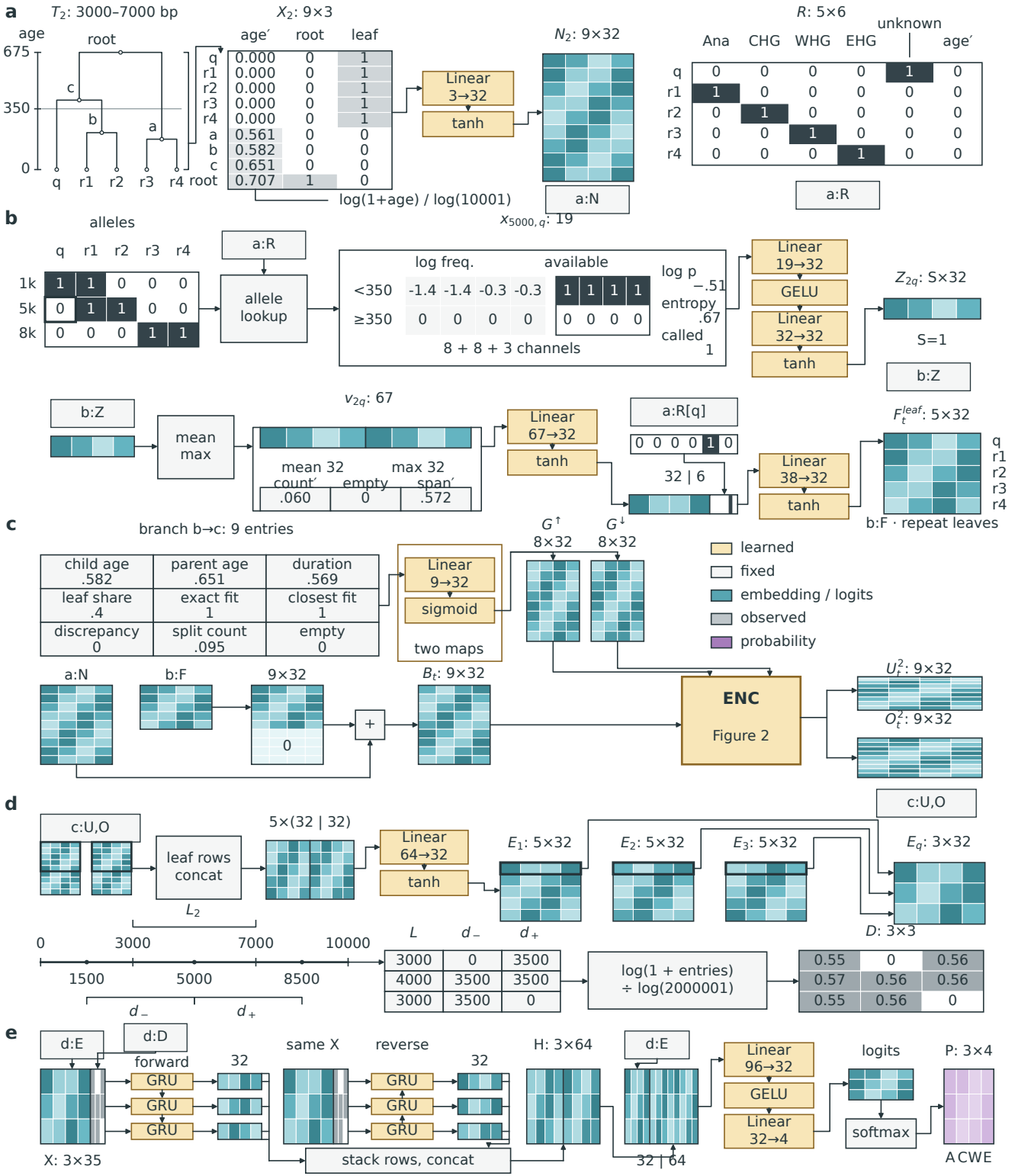

a
T_2: 3000–7000 bp
X_2: 9×3
N_2: 9×32
R: 5×6
age
675
350
0
root
c
b
a
q r1 r2 r3 r4
age´ root leaf
q 0.000 0 1
r1 0.000 0 1
r2 0.000 0 1
r3 0.000 0 1
r4 0.000 0 1
a 0.561 0 0
b 0.582 0 0
c 0.651 0 0
root 0.707 1 0
log(1+age) / log(10001)
Linear 3→32
tanh
a:N
$x_{5000,q}$: 19
Ana CHG WHG EHG unknown age´
q 0 0 0 0 1 0
r1 1 0 0 0 0 0
r2 0 1 0 0 0 0
r3 0 0 1 0 0 0
r4 0 0 0 1 0 0
a:R
b
alleles
q r1 r2 r3 r4
1k 1 1 0 0 0
5k 0 1 1 0 0
8k 0 0 0 1 1
a:R
allele lookup
log freq. available
<350 -1.4 -1.4 -0.3 -0.3 1 1 1 1
≥350 0 0 0 0 0 0 0 0
8 + 8 + 3 channels
log p −.51
entropy .67
called 1
Linear 19→32
GELU
Linear 32→32
tanh
Z_{2q}: S×32
S=1
b:Z
b:Z
mean max
v_{2q}: 67
mean 32 count´ .060
empty 0
max 32 span´ .572
Linear 67→32
tanh
a:R[q]
0 0 0 0 1 0
32 | 6
Linear 38→32
tanh
F_t^{leaf}: 5×32
q r1 r2 r3 r4
b:F · repeat leaves
c
branch b→c: 9 entries
child age .582
parent age .651
duration .569
leaf share .4
exact fit 1
closest fit 1
discrepancy 0
split count .095
empty 0
Linear 9→32
sigmoid
two maps
$G^{\uparrow}$ 8×32
$G^{\downarrow}$ 8×32
learned
fixed
embedding / logits
observed
probability
a:N
b:F
9×32
0
+
B_t: 9×32
ENC
Figure 2
U_t^2: 9×32
O_t^2: 9×32
d
c:U,O
leaf rows concat
5×(32 | 32)
L_2
Linear 64→32
tanh
E_1: 5×32
E_2: 5×32
E_3: 5×32
E_q: 3×32
c:U,O
D: 3×3
0 3000 7000 10000
1500 5000 8500
d_- d_+
L d_- d_+
3000 0 3500
4000 3500 3500
3000 3500 0
log(1 + entries) ÷ log(2000001)
0.55 0 0.56
0.57 0.56 0.56
0.55 0.56 0
e
d:E
d:D
forward
GRU
GRU
GRU
32
same X
reverse
GRU
GRU
GRU
32
stack rows, concat
X: 3×35
H: 3×64
d:E
32 | 64
Linear 96→32
GELU
Linear 32→4
logits
softmax
P: 3×4
ACWE